

From cells to donor-level prediction

One CMV label per donor

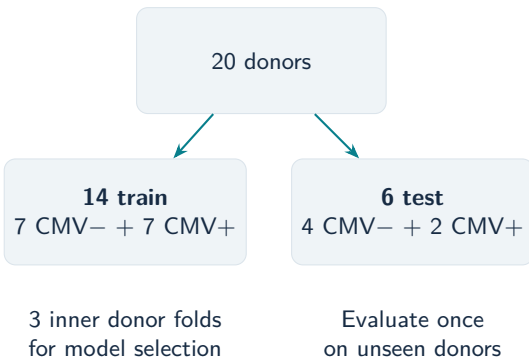
Donors	20
CMV− / +	11 / 9
Protein markers	37
Input gate	gated_alive

$$a_{ij} = \text{asinh}(x_{ij}/5)$$

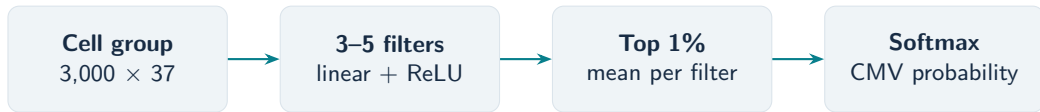
i : cell; j : marker; x_{ij} : raw value; a_{ij} : transformed value; 5: cofactor.

CellCNN/SVM: fit marker scaling on training donors only. Citrus uses ArcSinh values directly.

Horowitz et al. (2013), [doi:10.1126/scitranslmed.3006702](https://doi.org/10.1126/scitranslmed.3006702). Arvaniti & Claassen (2017), [doi:10.1038/ncomms14825](https://doi.org/10.1038/ncomms14825).



CellCNN learns filters for informative cells



Learn which cells matter

$$r_{if} = \max(0, \mathbf{w}_f^\top \mathbf{z}_i + b_f)$$

i : cell; f : filter; $\mathbf{z}_i$: 37 standardized markers; $\mathbf{w}_f$: filter weights; b_f : bias; r_{if} : response. $\mathbf{w}_f^\top \mathbf{z}_i$: weighted sum; $\max(0, \cdot)$: ReLU.

A filter scores a marker combination. Pooling focuses on the strongest responses.

Arvaniti & Claassen (2017), [doi:10.1038/ncomms14825](https://doi.org/10.1038/ncomms14825).

Our paper-guided implementation

- PyTorch; 200 cell groups per training donor.
- Choose the best inner-fold network.
- Adaptations: fixed 3/4/5-filter search; average five test-input probabilities.

Running the original Citrus implementation

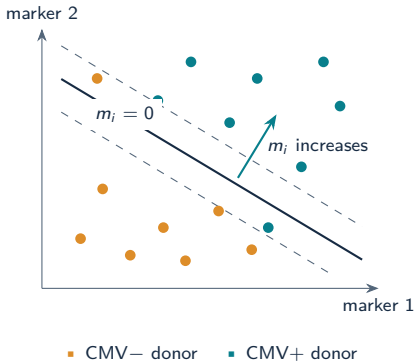


Two routes to a paper-guided implementation

CellCNN: new PyTorch implementation based on the paper and reference code.

Citrus: original legacy package code; recorded runtime: R 4.5.3.

A linear SVM assigns a score to each cell



Train on donor-labelled cells

10,000 cells per training donor; each cell receives its donor's label.

$$m_i = w^T z_i + b$$

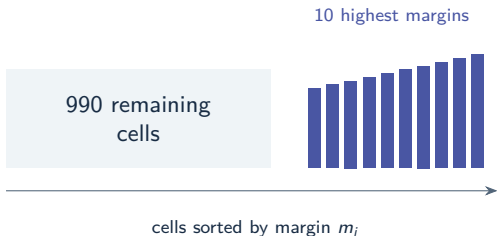
A larger margin means stronger evidence in the model's positive direction.

Linear SVM: L2 penalty and squared-hinge loss. The margin is a score, not a probability.

Arvaniti & Claassen (2017), [doi:10.1038/ncomms14825](https://doi.org/10.1038/ncomms14825).

Our adaptation: aggregate the highest cell margins

A few high-scoring cells can matter



Toy example: 1,000 cells; keep 10.

Margins 2.1, 2.2, $\dots$, 3.0 give $s_d = 2.55$.

At $\tau = 1.80$: predict CMV+.

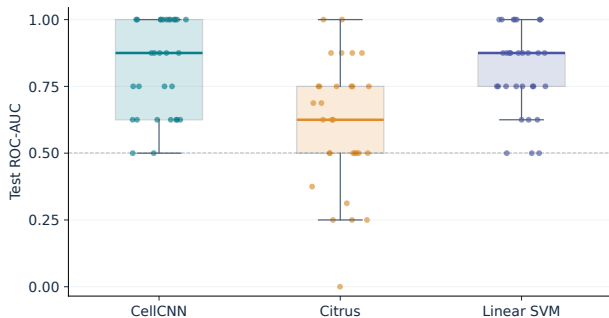
One score per donor

$$k_d = \lceil 0.01 N_d \rceil, \quad s_d = \frac{1}{k_d} \sum_{i \in I_d} m_i$$

d : donor; N_d : its cell count; k_d : count rounded up; I_d : its top-scoring cell indices; i : cell; m_i : cell margin; s_d : mean donor score; τ : donor threshold.

1. Select loss weight $C \in \{0.01, 0.1, 1\}$ by inner donor ROC-AUC.
2. Learn τ from inner out-of-fold donor scores.
3. Refit on 14 training donors; score all test cells.

Performance across 30 shared donor splits



Six test donors per split. Boxes show Q1–Q3; dots show individual splits.

BA thresholds: 0.5 for CellCNN/Citrus; inner-trained for SVM.

Median across 30 splits

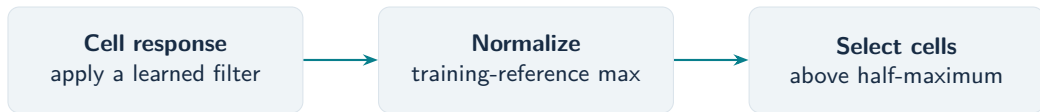
Method	AUC	AP	BA
CellCNN	0.875	0.833	0.625
Citrus	0.625	0.542	0.500
Linear SVM	0.875	0.833	0.625

AUC: ranking of CMV+ versus CMV−.

AP: average precision.

BA: balanced accuracy.

CellCNN: select cells with a strong filter response



One score per cell and filter

$$q_{if} = \frac{r_{if}}{M_f}, \quad q_{if} > 0.5$$

- i : cell
- f : filter
- r_{if} : filter response
- M_f : its positive maximum on the original training reference
- q_{if} : normalized score.

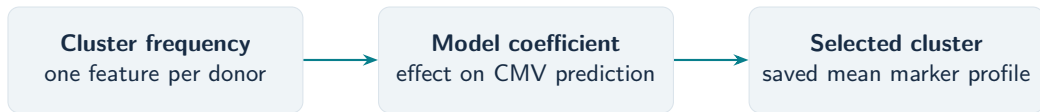
A simple example

Reference maximum: **10**. Selection threshold: **5**.

Responses **8 and 6** pass; responses **5 and 2** do not.

Use effective filters. Their output weights determine the CMV direction.

Citrus: select clusters used by the classifier



A score for a whole cluster

$$q_j = |\beta_j|, \quad q_j > 10^{-10}$$

j : cluster

β_j : its frequency coefficient in the selected logistic model

q_j : absolute coefficient.

The tolerance excludes numerical zeros.

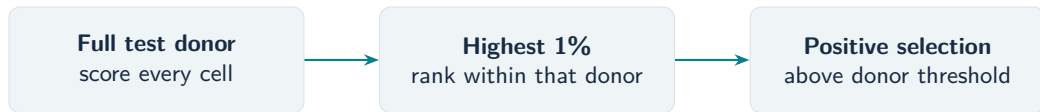
What this selects

All clusters with an effective coefficient enter the interpretation.

Positive coefficient: higher frequency supports CMV+; negative: CMV−, with other features fixed.

The stored centroid summarizes the cluster's training cells.

SVM: select the strongest positive cell scores



1. Compute the cell score

$$m_i = w^\top z_i + b$$

i : cell
 z_i : standardized marker vector
 w : learned weights
 b : intercept
 m_i : margin
 $w^\top z_i$: weighted sum.

2. Check the selected top cells

$$t_i = m_i - \tau, \quad t_i > 0$$

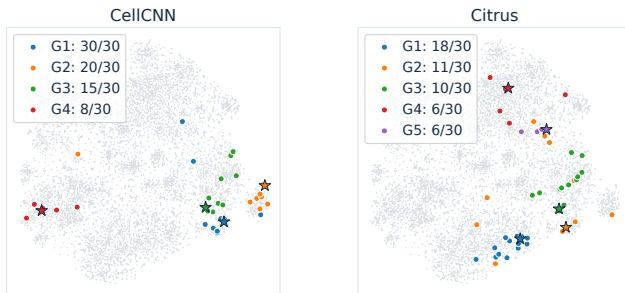
τ : donor threshold learned in inner validation
 t_i : cell score relative to that threshold.

A positive value passes **only within the highest 1%** of the full donor.

Which CellCNN and Citrus subsets recur?

Group similar subset centroids separately for each method; count distinct splits.

10,000 gated_alive cells; 500/donor; t-SNE perplexity 30; model splits 0–29.



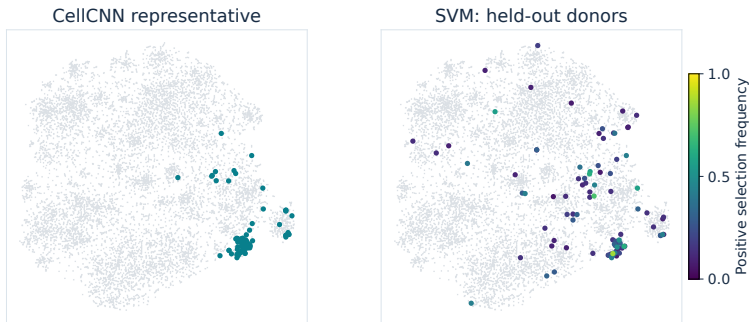
Show groups present in at least 6 of 30 splits (20%). Top groups: **30/30** and **18/30**, both positive.

Arvaniti & Claassen (2017), [doi:10.1038/ncomms14825](https://doi.org/10.1038/ncomms14825).

Which cells does the SVM repeatedly select?

Colour shows how often the same cell passes the positive SVM selection rule.

10,000 gated_alive cells; 500/donor; t-SNE perplexity 30; model splits 0–29.

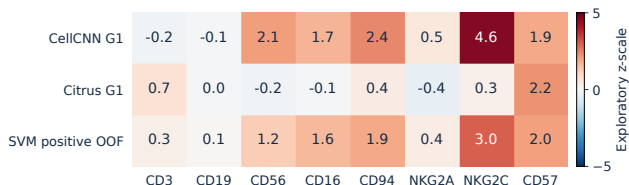


CellCNN: 87 map cells (exploratory). **SVM:** 93 map cells selected positively at least once.

Arvaniti & Claassen (2017), [doi:10.1038/ncomms14825](https://doi.org/10.1038/ncomms14825).

What do the marker profiles support?

CellCNN/Citrus: group representatives. SVM: positive test-cell profiles, with equal donor weights.



CellCNN

NKG2C/CD57 with CD56/CD16/CD94: compatible with a memory-like NK profile.

Citrus

Higher CD3/CD57 and much less NKG2C elevation suggest a T-associated pattern.

SVM positive OOF

NKG2C/CD57 and NK-associated markers are elevated; CD3 is also above the reference mean.

Arvaniti & Claassen (2017), [doi:10.1038/ncomms14825](https://doi.org/10.1038/ncomms14825). Horowitz et al. (2013), [doi:10.1126/scitranslmed.3006702](https://doi.org/10.1126/scitranslmed.3006702).

Backup: hyperparameters and paper deviations

	Implemented settings	Selection / paper relationship
CellCNN	3,000 cells/input; 200 inputs/donor; 3/4/5 filters; top 1%; Adam 0.01; batch 128; L2 10^{-4} ; no dropout; ≤ 100 epochs; patience 5.	Best validation accuracy; ties: AUC, loss, fewer filters, fold order. Fixed search; five rather than one test input. Selected net uses 9/10 donors.
Citrus	10,000 cells/donor; Ward; 37 ArcSinh markers; minimum cluster fraction 0.0005; L1 logistic; cv.min; no marker z-scoring.	10,000 cells/30 repeats versus paper's 20,000/100. Inner trees fitted per fold; original lambda grid uses all 14 outer training donors. Final fit on 14 donors.
SVM	10,000 cells/training donor; train-only scaler; L2 squared hinge; loss weight $C \in \{0.01, 0.1, 1\}$; top 1% of full-test margins (count rounded up).	Maximize mean inner AUC; ties: smaller C. Inner OOF threshold maximizes sensitivity + specificity. Final fit on 14 donors. Top-1-% aggregation is our adaptation.

Arvaniti & Claassen (2017), [doi:10.1038/ncomms14825](https://doi.org/10.1038/ncomms14825). Bruggner et al. (2014), [doi:10.1073/pnas.1408792111](https://doi.org/10.1073/pnas.1408792111).

Backup: SVM approach 1 — training the cell scorer

Training data and scaling

Sample 10,000 cells per training donor without replacement, with fixed seeds. Each cell inherits its donor's label. Fit a shared scaler on these training cells only.

$$a_{ij} = \text{asinh}(x_{ij}/5), \quad z_{ij} = \frac{a_{ij} - \mu_j}{\sigma_j}$$

i : cell; j : marker; x_{ij} : raw value; a_{ij} : transformed value; 5: cofactor; μ_j, σ_j : training mean and standard deviation; z_{ij} : standardized value.

Equal cell counts give donors equal weight in the training loss. No extra class weighting.

Labels provide weak cell supervision. The 37 markers yield 37 learned weights and one intercept.

Linear score and squared-hinge loss

$$m_i = \sum_{j=1}^{37} w_j z_{ij} + b$$

$$\min_{w,b} \left[\frac{1}{2} (\|w\|^2 + b^2) + C \sum_{i=1}^N \max(0, 1 - y_i m_i)^2 \right]$$

w : vector of marker weights w_j ; b : intercept; m_i : cell margin; N : training-cell count; y_i : donor label, -1 for CMV $-$ and $+1$ for CMV $+$.

C : loss weight; $\|w\|^2$: sum of squared weights; $\min_{w,b}$: optimize weights and intercept. The squared maximum is the squared-hinge loss.

Smaller C : more regularization. Larger C : more weight on fitting labels.

LinearSVC with `intercept_scaling=1` also penalizes b . No probability calibration.

Backup: SVM approach 2 — model selection and test

1. Choose settings within 14 donors

Use 3 inner donor folds. Test loss weights $C \in \{0.01, 0.1, 1\}$ on identical cell samples and scalers within each fold.

Highest mean inner donor ROC-AUC wins; ties favour smaller C . For that C , collect one out-of-fold score per training donor.

$$J(\tau) = \text{sens}(\tau) + \text{spec}(\tau) - 1$$

τ : candidate donor threshold; J : Youden index; **sens**: fraction of CMV+ donors predicted positive; **spec**: fraction of CMV− donors predicted negative.

Maximize J over finite ROC thresholds; ties choose the highest offered optimum.

2. Refit, then evaluate six test donors

Resample 10,000 cells from each of the 14 training donors; fit a new scaler and SVM. Keep the inner-selected threshold.

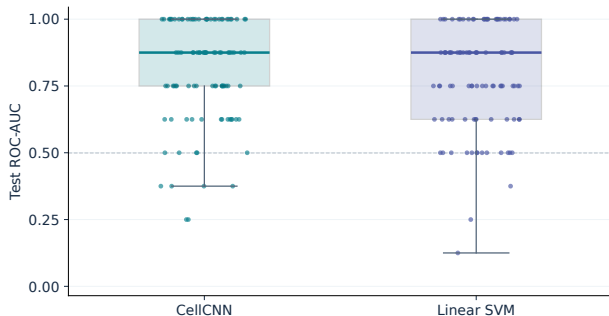
$$k_d = \lceil 0.01 N_d \rceil, \quad s_d = \frac{1}{k_d} \sum_{i \in I_d} m_i$$

d : donor; N_d : its full cell count; k_d : count rounded up; I_d : top-scoring cell indices; i : cell; m_i : margin; s_d : donor score.

Predict CMV+ when $s_d \geq \tau$. AUC uses continuous scores; balanced accuracy uses thresholded labels.

The uncalibrated threshold transfers from inner models to a refitted model. Changes in score scale can affect thresholded predictions. Outer test donors choose neither C nor τ .

Backup: CellCNN and SVM across 100 splits



Separate two-method comparison

Method	AUC	AP	BA
CellCNN	0.875	0.833	0.750
Linear SVM	0.875	0.833	0.625

Medians across splits 0–99.

AUC: ROC-AUC; AP: average precision;
BA: balanced accuracy.

600 held-out donor predictions per method, from the same 20 donors.

Both median AUCs remain 0.875. More splits do not add independent donors.

Boxes: Q1–Q3; dots: individual splits; dashed line: AUC 0.5.

Backup: CellCNN subset selection in detail

All quantities refer to one saved model and its original training reference.

Response and reference maximum

$$r_{if} = \max(0, \mathbf{w}_f^\top \mathbf{z}_i + b_f)$$

$$M_f = \max_{\ell \in T} r_{\ell f}, \quad q_{if} = r_{if} / M_f$$

i, ℓ : cells; f : filter; $\mathbf{z}_i$: 37 markers standardized by the saved scaler; $\mathbf{w}_f$: weights; b_f : bias; r_{if} : ReLU response; $\mathbf{w}_f^\top \mathbf{z}_i$: weighted sum.

T : reference cells; M_f : maximum response; q_{if} : normalized score, defined for $M_f > 0$.

Reconstruct **20,000 cells per actual training donor** with the original seed: 9 or 10 donors, hence 180,000 or 200,000 reference cells.

Direction, subset and centroid

$$\Delta_f = v_{f,+} - v_{f,-}$$

$$S_f = \{i \in T : q_{if} > 0.5\}, \quad c_f = \frac{1}{|S_f|} \sum_{i \in S_f} a_i$$

$v_{f,+}, v_{f,-}$: weights from filter f to the CMV+/CMV− outputs; Δ_f : output contrast. Its sign gives the direction.

S_f : selected reference cells; $|S_f|$: their count; a_i : ArcSinh marker vector; c_f : mean marker vector (centroid).

Skip filters with zero maximum or zero output contrast. Equality at half-maximum is excluded.

Selected cells have equal weight; donor contributions vary. New cells can score above 1.

Backup: Citrus subset selection in detail

Which frequency features matter?

$$\eta_d = \beta_0 + \sum_j \beta_j h_{dj}$$

d : donor; j : cluster; h_{dj} : cluster-frequency feature; β_0 : intercept; β_j : coefficient; η_d : linear predictor for CMV+.

`glmnet` standardizes frequency features internally.

$$q_j = |\beta_j| > 10^{-10}$$

q_j : absolute coefficient; 10^{-10} : numerical zero tolerance.

Retain every effective cluster. A positive coefficient raises the CMV+ predictor as frequency increases, with other features fixed; a negative coefficient lowers it.

What the exports preserve

$$c_j = \frac{1}{|A_j|} \sum_{i \in A_j} a_i$$

A_j : training cells belonging to cluster j ; $|A_j|$: their count; i : cell; a_i : ArcSinh marker vector; c_j : saved cluster centroid.

Training uses 10,000 cells/donor and retains clusters from 0.05% of training cells. Parent and child clusters can share cells.

Stored centroids and coefficients do not preserve the full tree and cell memberships needed to assign arbitrary map cells.

Assigning cells to the nearest centroid would introduce a different selection rule. Null models remain in the 30-split recurrence denominator.

Backup: SVM cell selection and recurrence

Exact selection within a test donor

$$k_d = \lceil 0.01 N_d \rceil, \quad t_{is} = m_{is} - \tau_s$$

d : donor; s : outer split; i : cell; N_d : full cell count; k_d : count rounded up; m_{is} : cell margin; τ_s : learned donor threshold; t_{is} : centered score.

Let I_{ds} contain the k_d highest-scoring cell indices. Break boundary ties by original FCS event number.

Within I_{ds} : positive if $t_{is} > 0$, negative if $t_{is} < 0$, neither if $t_{is} = 0$. Cells outside I_{ds} are unselected.

Example: 101 cells give 2 top cells. Margins 9 and 8 at threshold 8.5 give one positive and one negative selection.

Repeat across that donor's test visits

$$f_i^+ = \frac{n_i^+}{|\mathcal{T}_d|}$$

$\mathcal{T}_d$: all splits testing donor d ; $|\mathcal{T}_d|$: their count; n_i^+ : visits where cell i is in I_{ds} and $t_{is} > 0$; f_i^+ : positive selection frequency.

Include visits without selection: 3 positive selections in 8 test visits give 3/8.

$$s_{ds} - \tau_s = \frac{1}{k_d} \sum_{i \in I_{ds}} t_{is}$$

s_{ds} : mean of the selected cell margins, the donor score in split s .

Fixed-selection decomposition of the donor score; no causal cell-removal effect.

Backup: CellCNN — from cells to one plotted point

Slide 10: one selected model per split, one subset profile per effective filter

1. Reconstruct the training reference

20,000 cells per actual inner training donor; 9 or 10 donors give 180,000 or 200,000 reference cells. Reuse the original seed.

2. Select cells for each filter

Apply the saved scaler and filter. Keep responses strictly above half of that filter's reference maximum.

3. Average the selected marker profiles

Average each of the 37 ArcSinh-transformed markers across the selected cells. One filter gives one centroid, carrying its split and filter ID.

Actual example: split 9, filter 2

Reference cells	200,000
Half-maximum-selected cells	1,633
Result	1 centroid

This centroid becomes the representative of G1. The later cell map applies the same filter to a different, 10,000-cell display sample.

Mean, not median

Illustrative values for one marker: 1, 2, 9. Their mean is 4; their median is 2. We use the mean.

Across 30 selected models: 7 have 3 filters, 6 have 4, and 17 have 5. Together they yield **130 subset centroids**.

Backup: Citrus — two levels of clustering

Slide 10: original cell clusters first, groups of their profiles afterwards

1. Cluster cells within each split

The final Citrus model uses 10,000 cells from each of 14 training donors: **140,000 cells**.

A Ward tree defines cell clusters. L1-logistic regression uses their donor frequencies and retains effective coefficients of either sign.

For every selected cluster, average each ArcSinh marker over its training members. This gives **one centroid per selected cluster**.

Across 30 splits: **123 selected cluster centroids**. Three splits select no effective clusters; they remain in the recurrence denominator.

2. Group profiles across the splits

Compare the 123 centroids in the common standardized marker space. Similar profiles form **17 groups**; 5 occur in at least 6 splits and are shown.

Each coloured point still belongs to an original cluster in a specific split. A colour identifies its profile group.

Actual example: Citrus G1

28 cluster centroids from 18 splits.

Its star is one existing member: cluster 139891 from split 25. It is not an average of all 28 cluster profiles.

Backup: Colours, stars and overlapping centroids

Slide 10: grouping and display are separate operations

Colour: a group of similar profiles

Use all 37 markers on the shared exploratory z-scale. Group separately for each method with cosine distance, average linkage and a cut at 0.4.

Average linkage averages pairwise distances between groups. It does not replace their members with one mean profile.

Coefficient signs are ignored during grouping: a group may be positive, negative or mixed. Show groups present in at least **6 distinct splits**; each split counts once.

Star: an existing representative

Choose the member with the smallest summed cosine distance to its group members. This is the **medoid**, not a median or a new mean profile.

Position: the nearest map cell

Reuse the Task 2 map: 10,000 cells, 500/donor. Assign each centroid the t-SNE position of its nearest map cell in standardized marker space (Euclidean distance).

Several centroids can share a position. The star need not look central in the 2D map.

	All centroids	Shown groups	Plotted centroids	Distinct positions
CellCNN	130	4 of 15	111	44
Citrus	123	5 of 17	77	52

Backup: Individual cells and SVM selection frequency

Slide 11: each coloured point is one original cell in the 10,000-cell map

Left: one CellCNN representative

Use the G1 representative: **split 9, filter 2**. Apply its saved scaler and filter to all 10,000 map cells from all 20 donors.

Keep the original half-maximum threshold from its training reference; do not recompute a maximum on the map.

87 map cells pass and are coloured turquoise.

Colour indicates selection, not response strength or recurrence.

The centroid used 1,633 selected reference cells.

The 87 points are a separate application to the map sample. This is exploratory and includes training donors.

Right: repeated SVM test evaluations

Each saved SVM scores only its outer test donors. Select the highest 1% of each **full donor's** margins; positive selection also requires exceeding the learned donor threshold.

Count positive selections of the same map cell across **all test appearances of its donor**: 4–16 appearances, including those without selection.

Example: 3 positive selections in 8 test appearances give a frequency of **0.375**. Plot that cell once with this colour.

93 map cells pass at least once. Zero frequency stays grey; brighter colours mean more consistent positive selection.

Backup: Which cells enter each heatmap row?

Slide 12: eight displayed markers; all profiles start on the ArcSinh scale

Row	Cells used	How the profile is formed
CellCNN G1	Split 9/filter 2: 1,633 of 200,000 reference cells; 10 inner training donors.	Mean markers of one G1 representative; not all G1 centroids or the 87 map cells.
Citrus G1	Split 25/cluster 139891: its members within 140,000 training cells from 14 donors.	Mean markers of one G1 representative cluster; not all 28 G1 centroids.
SVM positive OOF	Positively selected cells from full outer test donors in splits 0–29; 168 of 180 visits are nonempty.	Average cells within visits; nonempty visits within donors; then the 20 donors equally.

Positive OOF

Top-1-% cells above the learned threshold, scored only when their donor was held out. Both CMV classes can contribute.

Equal donor weights

Each donor contributes equally, regardless of cell count or test visits. Use full-donor selections, not just the 93 map cells.

Backup: Reading the shared marker scale

Slide 12: expression relative to a common reference, not classifier weights

Standardize each mean marker value

$$z_j = \frac{c_j - \mu_j}{\sigma_j}$$

j : marker
 c_j : profile's mean ArcSinh marker value
 μ_j : reference mean of that marker
 σ_j : reference standard deviation
 z_j : value shown in the heatmap

The shared reference

10,000 map cells, 500 per donor, including both CMV classes. Reuse the stored exploratory scaler from Task 2 for every heatmap row.

The model-specific scalers still govern cell selection; this shared scale is used for display.

Read the number and colour together

Value	Relative to the reference mean
-1	1 standard deviation below
0	Equal to the reference mean
+1	1 standard deviation above

Example: CellCNN NKG2C = 4.6

4.6 reference standard deviations above the reference mean; not 4.6 times the expression.

Red: above reference. Blue: below. These signs do not describe CMV class or model direction. Zero means reference-level expression.

Mean profiles can hide heterogeneous cells. These values are not cell-type gates.